Bioresource Technology Reports 27 (2024) 101949



Contents lists available at ScienceDirect

Bioresource Technology Reports

journal homepage: www.sciencedirect.com/journal/bioresource-technology-reports





Microbial community dynamics and volatile fatty acid production during anaerobic digestion of microaerated food waste under different organic loadings

Farrukh Raza Amin [a,b,c,1], Habiba Khalid [a,b,d,1], Jingjing Wang [a,b], Yaxiang Li [a,b], Longxue Ma [a,b], Wuxi Chen [a,b], Yu Duan [a,b], Yida Zhai [a,b], Demao Li [a,b,*]

[a] Tianjin Institute of Industrial Biotechnology, Chinese Academy of Sciences, Tianjin 300308, China
[b] National Innovation Centre for Synthetic Biology, Tianjin 300308, China
[c] Department of Chemistry, COMSATS University Islamabad, Park Road, Tarlai Kalan 45550, Islamabad, Pakistan
[d] Department of Applied Bioeconomy, Wrocław University of Environmental and Life Sciences, 50-375 Wrocław, Poland

ARTICLE INFO

Keywords:
Food waste
Oxygen pretreatment
Organic loading
Volatile fatty acids
Microbial community analysis

ABSTRACT

This study examined the effects of microaeration pretreatment and varying organic loading rates (OLRs) on volatile fatty acids (VFAs) production from food waste (FW) *via* anaerobic digestion process. FW was pretreated with 6 mL O_2/g VS and digested at OLRs ranging from 10 to 25 g/L VS. Microaeration pretreatment increased the VFA yield to 7.4 g/L, whereas without pretreatment, it was 5.19 g/L, with butyrate as the dominant VFA (92 % at 25 g/L OLR) due to *Firmicutes* enrichment. A relatively high OLR initially increased VFA production but later favored propionate without pretreatment. Microbial analysis revealed that aerobic conditions promoted lactic acid bacteria such as *Lactobacillus*, increasing lactic acid production, whereas anaerobic conditions promoted *Prevotella* and butyrate producers such as *Clostridium*, which specialize in acetate and butyrate production. This study highlights that microaeration and an optimal OLR can increase and stabilize VFA yields, modulating the microbial community for tailored VFA profiles that are useful in biofuel and biochemical applications.

1. Introduction

The midterm reorientation of waste management is anticipated to be due to the current change in waste policy within the circular economy. A significant improvement over the previous regulations is the inclusion of specific measures to manage waste and encourage the reuse of materials. Moreover, the perception of waste has shifted from being an undesirable product that can be costly to handle and dispose of (Aguirre-Garcia, 2019).

Food waste (FW) is produced in factories and kitchens and consists of both leftover cooked food and raw ingredients. FW is the main component of municipal solid waste (MSW), which accounts for 37–62 % of MSW in China and 12–30 % of MSW in certain developed countries, such as the USA, Singapore, and England (Wang et al., 2020). Although 870 million people are chronically undernourished, 1.3 billion to 1.6 billion tons of food waste constitute a quarter of global food production (Braguglia et al., 2018). The level of social and economic development in an area has a significant effect on the quantity of FW. Europe and North America have FW yields per capita of approximately 0.26–0.32 kg/d, whereas Africa and Southeast Asia have yields of only 0.02–0.053 kg/d (Bao et al., 2015). By 2025, Asia is expected to experience an increase in FW from 2.78 billion tons to 4.16 billion tons. The annual growth rate of FW has surpassed 10 % in recent years, especially in China, as a result of the acceleration of industrial development and

Abbreviations: AD, anaerobic digestion; ANOVA, analysis of variance; COD, chemical oxygen demand; FID, flame ionization detector; FW, food waste; H_2, hydrogen; HRT, hydraulic retention time; MSW, municipal solid waste; N, nitrogen; NH_4^+-N, ammonium nitrogen; OLR, organic loading rate; OTUs, operational taxonomic units; PCR, the polymerase chain reaction; PS, pig slurry; RA, relative abundance; RDP, ribosomal database project; SCCA, short-chain carboxylic acids; TS, total solids; VFA, volatile fatty acids; VS, volatile solids; WOT, without treatment; WT, with treatment.

* Corresponding author at: Tianjin Institute of Industrial Biotechnology, Chinese Academy of Sciences, 32 West 7th Avenue, Tianjin Airport Economic Area, Tianjin 300308, China.
E-mail address: li_dm@tib.cas.cn (D. Li).
[1] These authors contributed equally.

https://doi.org/10.1016/j.biteb.2024.101949
Received 13 May 2024; Received in revised form 27 August 2024; Accepted 28 August 2024
Available online 30 August 2024
2589-014X/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

urbanization (Ren et al., 2018).

The expanding worldwide population has led to a surge in the number of FW in need of adequate pretreatment. Compared with other pretreatment methods, such as landfilling, incineration, and composting, FW acidogenic fermentation for VFAs has gained widespread acceptance as a versatile and environmentally friendly method. On the other hand, short-chain carboxylic acids (SCCA), such as acetic acid, propionic acid, and butyric acid, are valuable platform chemicals generated during acidogenic processes (Bhatt et al., 2020). With a price range of 2000–2500 USD/t, butyric acid is considered an important chemical product among SCCAs and is significantly more expensive than acetic acid is (400–900 USD/t) (Bastidas-Oyanedel and Schmidt, 2019). Moreover, butyric acid can be used in a variety of ways as an intermediary or feedstock in the food, pharmaceutical, and chemical industries and thus has considerable commercial potential (Jiang et al., 2018). This process not only stabilizes these wastes but also offers an opportunity for chemical and energy recovery (Wainaina et al., 2019). The widely implemented process of anaerobic digestion (AD) is used to obtain value-added chemicals and methane from FW (Tenci et al., 2023). Microbial communities naturally decompose organic matter into methane-rich biogas through different stages, which can then be used for heating, electricity, or fuel. Research has focused mostly on the use of anaerobic digestion to produce useful byproducts other than methane, such as H_2 and carboxylic acids, including lactic acid, succinic acid, and VFAs (Kleerebezem et al., 2015; Selder et al., 2021; Uddin and Wright, 2023).

The enzymatic hydrolysis of FW needs to be improved to produce more VFAs. Pretreating FW via physical, chemical, and biological pretreatments is a common practice. Physical procedures are used to decrease the substrate's particle size, whereas chemical pretreatment makes use of chemicals to increase biomass digestibility. However, these pretreatments are energy intensive and need to be handled carefully (Amin et al., 2017; Viancelli et al., 2023). The incorporation of microaeration pretreatment and enzymatic hydrolysis enhances the efficiency of VFAs production. This pretreatment process involves microorganisms being partially exposed to oxygen under moderate operating conditions. This process accelerates the hydrolysis stage by encouraging microbial activity and cell growth (Wagner et al., 2018). The conversion of FW into valuable VFAs is more efficient with the integration of microaeration pretreatment, which reduces the lag phase time and increases the maximum VFAs production rate. By adopting this holistic approach, we can efficiently utilize FW to generate VFAs for various biofuel or chemical production applications (Amin et al., 2021b; Ladero et al., 2018). The potential role of oxygen in regulating the organization of the microbial community makes it difficult to reconcile the beneficial and detrimental effects of microaeration during AD. A thorough examination of the underlying processes is also needed to obtain a deeper understanding of the microaeration pretreatment procedure.

This study chose a microbial reagent from pig slurry (PS), which is known to be essential for hydrolyzing FW and has diverse combinations of beneficial microbes from various development conditions, to perform microaeration pretreatment. The role of facultative bacteria in microaeration pretreatment is significant, as it is a biological process. The growth rate of facultative bacteria increases under microaerobic conditions, resulting in the production of fewer extracellular enzymes, such as cellulase and protease, which ultimately leads to a more efficient hydrolysis process. In Fu, the increase in *Firmicutes* abundance resulted in increased hydrolysis under microaerated conditions (Fu et al., 2016; Wang et al., 2024). The primary goals of this research are to (1) create a sustainable pretreatment technique that increases the production of VFAs in FW; (2) examine the effects of variations in organic loading in the fed-batch process of AD on VFAs production; and (3) look into the changes that occur in microbial communities throughout the AD process and how functional microorganisms react to different oxygen concentrations.

2. Materials and methods

2.1. Feedstock and inoculum

FW was collected from a canteen located at the Tianjin Institute of Industrial Biotechnology, Tianjin, China, and ground (to homogenize) with a high-speed grinder (XINGSHILIHE, Beijing, China) prior to being added to the fermentation system. PS was collected from farms in Hainan Dongzhaigang, China, and used as the startup inoculum for the AD process.

2.2. Oxygen pretreatment

One-liter bottles were used for oxygen pretreatment. A total volume of 100 mL was formed by mixing 10 g of wet food waste, 20 mL of PS, and deionized water during pretreatment. Then, the bottles were flushed with N_2 for 3 min to exhaust all the air, and the bottles were closed with a rubber stopper. After removing all the air from the bottles, 48 mL of oxygen was added to each bottle at atmospheric pressure *via* a syringe to achieve an oxygen level of 6 mL/g VS (marked as O_2 pretreated). We obtained this information from our previous study, where we optimized the oxygen load (Amin et al., 2021b). The process of oxygen pretreatment was performed in duplicate, and two bottles were placed without oxygen pretreatment as a control (marked without oxygen pretreatment). All the bottles were placed in a water bath at 37 °C and mixed with a magnetic stirrer during oxygen pretreatment.

2.3. Organic loading

Two identical reactors equipped with mixers were operated in fed-batch mode, employing both feeding and draw-off methods. The experiments were initiated under controlled conditions, maintaining a constant temperature of 37 °C and a neutral pH of 7.0 (set at the start of the experiment). The hydraulic retention time (HRT) was adjusted on the basis of the depletion of VFAs concentrations, with HRT values set at 9, 31, and 72 days. The impact of the organuc loading rate (OLR) on the process was explored by operating the reactors at OLRs of 10, 15, 20, and 25 g/L on a VS basis. The experimental period spanned 125 days, aiming to determine the optimal OLR for maximum production of each VFA.

2.4. Analytical methods

The FW and PS were analyzed *via* standard methods to determine the total solids (TS) and volatile solids (VS) (APHA, 2005). The weights of the FW and PS were recorded, and the samples were placed in an oven at 105 °C for 24 h to eliminate moisture. To calculate the TS content, as shown in Eq. (1), the FW and PS were reweighed after 24 h. The second step involved converting the dried samples into ash by heating them in a muffle furnace for 2 h at 550 °C. The samples were subsequently removed from the furnace and cooled to room temperature before being weighed again. Eq. (2) shows the calculation of the VS content.

$$TS = \frac{m_3 - m_1}{m_2 - m_1} \times 100 \quad (1)$$

$$VS = \frac{m_3 - m_4}{m_2 - m_1} \times 100 \quad (2)$$

where m_1 is the weight of the empty crucible, m_2 is the weight of the crucible and feedstock before drying, m_3 is the weight of the crucible and feedstock after drying, and m_4 is the weight of the crucible and ash after the muffle furnace. An elemental analyzer was used for elemental analysis (C, H, N, S, and O). By assuming that C, H, O, and N are 99.5 % under a VS assumption, C, H, O, and N were used to determine the oxygen content of the FW (Zhang et al., 2018). An automatic pH meter (AquaSearcherTM, OHAUS, China), which is designed for industrial

applications, was used to measure the pH. VFAs were determined by using a GC-7900 instrument (Techcomp) with a DB-WAXETR capillary column and flame ionization detector (FID) during operation at 60 °C and 250 °C. HACH test kits (HACH, USA) were used to measure chemical oxygen demand (COD) and ammonium nitrogen in accordance with the manufacturer's instructions.

2.5. Anaerobic digestion (AD)

AD was conducted in duplicate. With a feedstock-to-inoculum ratio of 1:1, the pretreated FW was inoculated with concentrated PS following the complete consumption of oxygen. The working volume was 0.8 mL with the use of deionized water. To ensure anaerobic conditions, the digesters were refilled with high-purity nitrogen, sealed, and incubated in a 37 °C water bath for 125 days. The agitation rate for all the digesters was maintained at 100 rpm. Samples were collected daily to analyze the concentration of VFAs. Fig. 1 shows a schematic illustration of the experimental setup.

2.6. Microbial community analysis

Two different stages of digestion were used to extract the genomic DNA of bacteria: on the 5th day and on the 97th day. The V3-V4 region of the 16S rRNA gene was amplified via the primers 338F (5'-ACTCC-TACGGGAGGCAGCAGAG-3′) and 806R (5'-GGACTACHGGTGTWTC-TAAT-3′). The polymerase chain reaction (PCR) was performed with initial denaturation at 95 °C for 3 min, followed by 25 cycles of 95 °C for 30 s, 55 °C for 30 s, and 72 °C for 30 s, with a final extension at 72 °C for 5 min. Sequencing was conducted on an Illumina MiSeq platform *via* 2 × 300 bp paired-end sequencing chemistry (Majorbio Bio-Pharm Technology Co., Ltd., Shanghai, China). To produce precisely tailored sequences, the sequencing data from MiSeq amplicon sequencing were spliced and checked for quality, and the adapters were removed. The raw sequences were processed *via* QIIME2 (version 2020.8). The Ribosomal Database Project (RDP) classifier was used to categorize the operational taxonomic units (OTUs) by clustering and taxonomically classifying them (Wang and Cole, 2024). Diversity analyses were conducted *via* the phyloseq R package. Dividing the number of sequences by the total number of sequences associated with a taxon was used to calculate the relative abundance (RA) (Mao et al., 2019). MOTHUR tools and Chao1 and ACE estimators were used to calculate the Shannon index for species diversity estimation (Li et al., 2018).

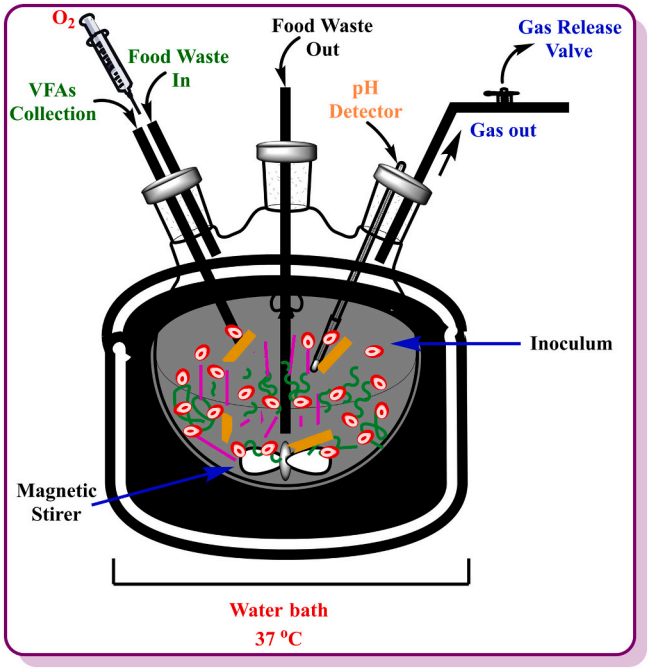


Fig. 1. Schematic diagram illustrating the experimental design.

2.7. Statistical analysis

All experiments were conducted in duplicate, and the results are presented as the mean values. Statistical analyses were performed *via* Origin Pro 9.0 (Origin Lab, USA). To identify significant differences in the experimental data, all the data were subjected to analysis of variance (ANOVA) and multiple comparisons; a P value of <0.05 was considered statistically significant. The diversity indices (Shannon, ACE, and Chao1) were calculated *via* the vegan package in R (version 4.0.3). Graphs and figures were generated *via* Origin Pro 9.0.

3. Results and discussion

3.1. Characteristics of FW and inoculum

Table 1 outlines the characteristics of the FW and inoculum used in the study. The FW contained 21.8 and 19.8 % TS and VS, respectively, whereas the inoculum contained only 5.2 % and 2.3 % TS and VS, respectively. The FW had a 91.0 % higher VS/TS ratio than did the inoculum, suggesting that it has a rich organic composition that is conducive to producing VFAs through AD. The pH values of the FW and inoculum were initially 4.8 and 8.5, but they were adjusted to 7.1 to ensure optimal AD conditions. The results of the nutrient content analysis revealed that the N_2 content in the FW was 3.0 %, which was slightly greater than the N_2 content in the inoculum, which was 2.7 %. Compared with the inoculum, the FW resulted in a significant increase in the C content of 47.7 % (26.5 %). The FW had H_2 and O_2 contents of 7.45 % and 32.4 %, respectively, whereas the inoculum had H_2 contents of 3.6 % and O_2 contents of 12.2 %. The S content in both the FW and inoculum was negligible. The FW and inoculum each had C/N ratios of 15.9 and 9.8, respectively, which are crucial parameters for AD process. This ratio shows that there is a balanced nutrient composition that facilitates the effective production of VFAs. Owing to its high organic content and appropriate nutrient profile, FW is capable of producing VFAs through AD, as demonstrated by this detailed analysis.

3.2. Enhancing VFA Production from Food Waste

3.2.1. Impact of the oxygen pretreatment and organic loading rates

By stopping the degradation process at the acidification stage, it is possible to modify the conventional AD process to favor the production of VFAs instead of methane. To effectively inhibit methanogens, several strategies have been explored, including chemical inhibitors, oxygen introduction, and substrate loading (Eryildiz and Taherzadeh, 2020). This investigation focused on VFAs, which are the primary products of the organic matter acidogenesis process, because of their critical role in

Table 1
Characteristics of food waste and inoculum.

Parameter	Food waste	Inoculum
TS (%)[a]	21.8 ± 0.2	5.2 ± 0.1
VS (%)[a]	19.8 ± 0.1	2.3 ± 0.0
VS/TS (%)	91.0	45.2
$pH_{initial}$	4.8	8.5
$pH_{adjusted}$	7.1	7.1
N (%)[b]	3.0 ± 0.1	2.7 ± 0.0
C (%)[b]	47.7 ± 0.1	26.5 ± 0.1
H (%)[b]	7.5 ± 0.0	3.6 ± 0.0
O (%)[b]	32.4 ± 0.2	12.2 ± 0.1
S (%)[b]	0 ± 0.0	0 ± 0.0
C/N	15.9	9.8

[a] Total weight of the sample, [b]TS of the sample.

hydrolysis and fermentation, which provides valuable insights into the efficiency of the process (Fu et al., 2015; Shen et al., 2017). This study presents data on the composition and concentration of VFAs produced from FW under different conditions, specifically comparing scenarios with and without oxygen pretreatment. The data contained in Table 2 show the VFAs yield at the point of maximum production. Furthermore, Fig. 2 shows the dynamic changes in the VFAs concentration and composition that occur with and without oxygen pretreatment at various OLRs. The concentration of VFAs initially increased in both reactors at an OLR of 10 g/L; it peaked on the 5th day in the reactor without oxygen pretreatment and on the 8th day in the reactor with oxygen pretreatment. On the other hand, after the 7th day, the VFAs concentration in the reactor without pretreatment decreased, but the VFAs levels in the reactor pretreated with oxygen remained constant. This observation led to an increase in the OLR to 15 g/L in both reactors, which further increased VFAs production. In the reactor without oxygen pretreatment, the highest VFAs generation occurred on day 12, whereas in the reactor with oxygen pretreatment, it occurred on day 13. After that, only the reactor without oxygen pretreatment presented a decrease in the VFAs concentration, which led to an increase in the OLR from 15 g/L to 20 g/L, leading to possible improvements in VFAs generation. On the 33rd day, the reactor with no oxygen pretreatment presented the highest VFAs output. On the other hand, the oxygen-treated reactor demonstrated steady VFAs generation, with the greatest output recorded on the 31st day, whereas a decreasing trend was detected from the 33rd to the 70th day. This trend led to a further increase in the OLR to 25 g/L, which, on the 92nd day in the reactor without oxygen pretreatment, produced the greatest amount of VFAs and was stable until the 125th day. In contrast, the highest VFAs production in the oxygen-treated reactor was noted on the 96th day, and VFAs production remained stable throughout the study period. In the fermentation process, the start-up phase was characterized by a gradual increase in the OLR to achieve a balance between alkalinity and acidification. The fermentation conducted at 37 °C had distinct outcomes on the basis of the oxygen availability in the reactors. In reactors without oxygen, butyrate emerged as the dominant VFA, constituting 81.3 vol% of all products at an OLR of 10 g/L, followed by acetate (9.4 vol%) and propionate (9.2 vol %). Notably, ethanol was absent under these conditions. Conversely, in the oxygen-treated reactors, the butyrate concentration increased to 91.5 vol%, whereas the propionate and acetate concentrations decreased significantly to 7.1 and 1.0 vol%, respectively. This trend of butyrate dominance persisted across higher OLRs (15, 20, and 25 g/L) in both the oxygen- and non-oxygen-treated reactors. Without oxygen pretreatment, the percentage of butyrate slightly decreased with increasing OLR (74.3, 70.5, and 69.4 vol%, respectively), whereas the percentage of propionate increased to 26 vol%. In contrast, the oxygen-treated reactors maintained or increased butyrate dominance (91.5, 85.5, 75.9, and 92.0 vol% across the OLRs tested), as shown in Table 2. These findings indicate that oxygen availability significantly ($P < 0.05$) influences VFA production during fermentation, with oxygen pretreatment favoring butyrate production. These findings suggest potential strategies for optimizing VFAs production through oxygen management in fermentation processes. The observed trends in butyrate production, in addition to the increasing concentrations of propionate during fermentation, suggest a potential shift in microbial populations favoring butyric acid-producing microorganisms, such as *Clostridium* species. This hypothesis aligns with previous findings by Komemoto et al. (2009), who reported high butyrate concentrations of 5600–6000 mg/L at 37 °C (Bhatt et al., 2020; Komemoto et al., 2009). Additionally, the study revealed optimal VFAs production at pH values ranging from 6.6 to 4.4, suggesting that hydrolytic enzyme activities thrive within this pH range (Fonseca et al., 2020), further increasing acid production, which is similar to the findings of Jiang et al. (Jiang et al., 2013). The variation in VFA content due to oxygen pretreatment reflects, to some extent, whether this technology could increase hydrolysis to yield more soluble small-molecule substances for acid production (Eryildiz and Taherzadeh, 2020). As shown in Fig. 2, oxygen pretreatment with the inoculum agent resulted in greater VFAs production than did pretreatment without oxygen. Overall, this approach not only increases the yield and stability of VFAs but also shifts the fermentation product spectrum, offering promising avenues for bioenergy and biochemical production from organic waste (Wang et al., 2024).

Notably, there were notable alterations in the VFA composition, even though the data in Table 2 indicate that the impact of oxygen pretreatment on overall VFA generation was rather minor. The distribution of individual VFAs was significantly affected by oxygen pretreatment, even if the total VFA yields were similar for pretreated and nonpretreated samples (e.g., 7.1 g/L for both the OLR 15 and the 7.4 g/L vs. 7.8 g/L at the OLR 25). For example, compared with the nonpretreated sample, the pretreated sample demonstrated a significant increase in butyrate production (92.0 vol% vs. 69.4 vol%) and a decrease in propionate production (3.5 vol% vs. 24.9 vol%) at an OLR of 25. The main impact of oxygen pretreatment appears to be this change in the VFAs profile rather than the total VFAs yield. The variability of microbial communities and the complexity of food waste may be the reasons for the comparable total VFA outputs. Pretreatment might have improved some metabolic pathways, especially those that support the formation of butyrate, but it is also likely that other factors, such as the availability of substrate or the rates at which microbes grow, limited the total amount of VFAs that could be produced. These results emphasize the importance of considering both the composition and total yield of VFAs when assessing pretreatment effects. Even if overall yields remain the same, the capacity to modify VFAs profiles through oxygen pretreatment may be useful for customizing VFAs production for certain downstream uses.

To further evaluate the efficiency of the VFAs production process, we calculated the VFAs yield and productivity for each condition. The VFAs yield, expressed as (gCOD/L VFA)/(gCOD/L fed), provides insight into the conversion efficiency of the substrate to VFAs. The VFAs productivity, expressed as gCOD/(L × day), indicates the rate of VFAs production. The results are presented in Table 3. The VFAs yield decreased with increasing OLR for both oxygen pretreated and untreated samples, suggesting that higher OLRs may lead to less efficient conversion of the substrate to VFAs. The highest yield was observed at the lowest OLR (10 gVS/L) for both conditions. VFAs productivity was generally greater at

Table 2
Effects of microaeration on VFA production from food waste at different OLRs.

	Without oxygen pretreatment				With oxygen pretreatment			
	OLR 10	OLR 15	OLR 20	OLR 25	OLR 10	OLR 15	OLR 20	OLR 25
VFAs (g/L)	5.19	7.1	7.7	7.8	4.8	7.1	5.0	7.4
Ethanol (%)*	0.0	0.7	0.1	0.0	0.0	0.6	1.3	1.9
Acetate (%)*	9.4	3.4	3.4	5.7	1.0	3.0	7.0	2.5
Propionate (%)*	9.2	21.6	25.9	24.9	7.5	10.9	15.8	3.5
Butyrate (%)*	81.3	74.3	70.5	69.4	91.5	85.5	75.9	92.0
pH	6.6	5.8	5.7	5.4	6.5	5.4	5.0	4.4
Day	5	12	33	92	8	13	31	96

*Day: highest VFAs observed at different OLRs, *%: volumetric concentrations.
A *t*-test, $P < 0.05$, revealed a statistically significant difference in VFA production at various OLRs with and without O_2 pretreatment.

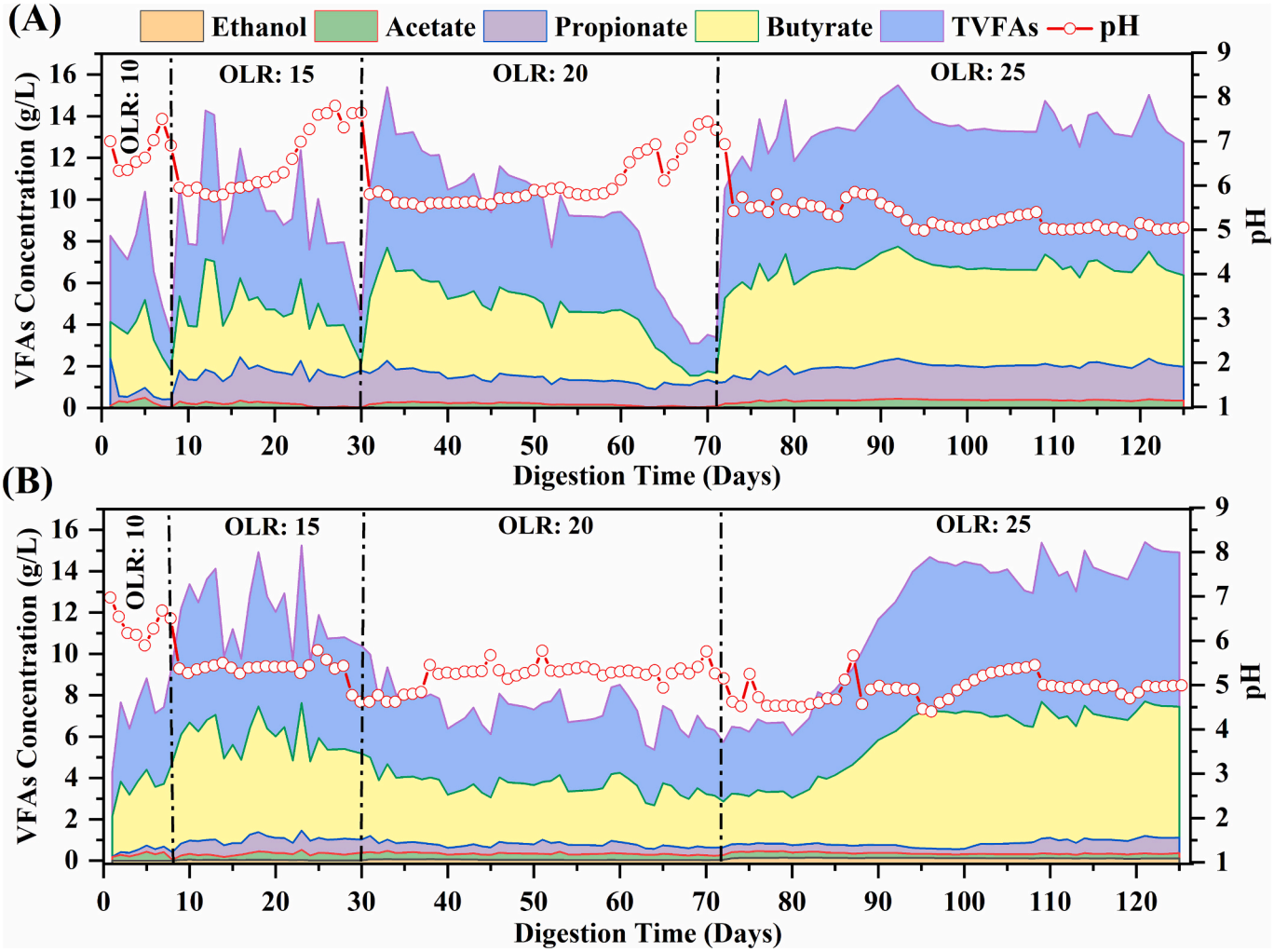


Fig. 2. VFA production of food waste: (A) without oxygen pretreatment, (B) with oxygen pretreatment.

Table 3
VFA yield and productivity at different OLRs with and without oxygen pretreatment.

OLR (gVS/L)	WOT (Yield) gCOD/L	WOT (Productivity) gCOD/L × d	WT (Yield) gCOD/L	WT (Productivity) gCOD/L × d
10	0.42	0.87	0.39	0.51
15	0.38	0.50	0.38	0.46
20	0.31	0.20	0.20	0.14
25	0.25	0.07	0.24	0.06

OLR: organic loading rate; WOT: without oxygen pretreatment; WT: with oxygen pretreatment.

lower OLRs and decreased as the OLR increased. This trend indicates that while higher OLRs provided more substrate, the microbial community's ability to convert this substrate to VFAs became limited, possibly due to inhibition or rate-limiting steps in the fermentation process. Interestingly, oxygen pretreatment did not consistently improve the VFAs yield or productivity across all the OLRs. This observation aligns with our earlier discussion on the similar total VFAs production between samples pretreated with oxygen and those not pretreated with oxygen, further emphasizing that the primary effect of oxygen pretreatment was on VFAs composition rather than overall yield or productivity.

3.3. Ammonia nitrogen and COD analysis at different OLRs

The fluctuations in the NH_4^+-N and COD concentrations in the hydrolysate with different OLRs with and without oxygen pretreatment are illustrated in Fig. 3. Fig. 3 (A) shows the NH_4^+-N concentration data variability across different phases of the experiment, reflecting how the microbial community responds to changes in OLR without oxygen pretreatment. The initial phase (days 1–8) with a 10 g/L OLR showed a gradual increase in NH_4^+-N levels, indicating initial adaptation. Increasing the OLR to 15 g/L (days 9–30) and subsequently to 20 g/L (days 31–71) resulted in fluctuating NH_4^+-N concentrations, suggesting that a stressed microbial environment struggled with nitrogen assimilation. The final phase (days 72–125), with a 25 g/L OLR, presented a more stable but elevated NH_4^+-N concentration (days 90–100), possibly indicating a balance between nitrogen release and assimilation. Ammonia is generated during anaerobic fermentation when nitrogenous compounds, particularly proteins, are degraded biologically and hydrolyzed into amino acids, which are then further degraded into ammonia (Jiang et al., 2013; Jun et al., 2009). Proteins are more easily hydrolyzed under neutral conditions, leading to a higher concentration of NH_4^+-N. The low amounts may be explained by a number of variables, including the possibility that some of the ammonia nitrogen was transformed during digestion into VFAs or other nitrogenous chemicals and that a large portion was ingested by the microbial population for development and metabolism (Wang et al., 2015). As previously mentioned, fermentation of FW at pH values of 6.0 and 7.0 resulted in

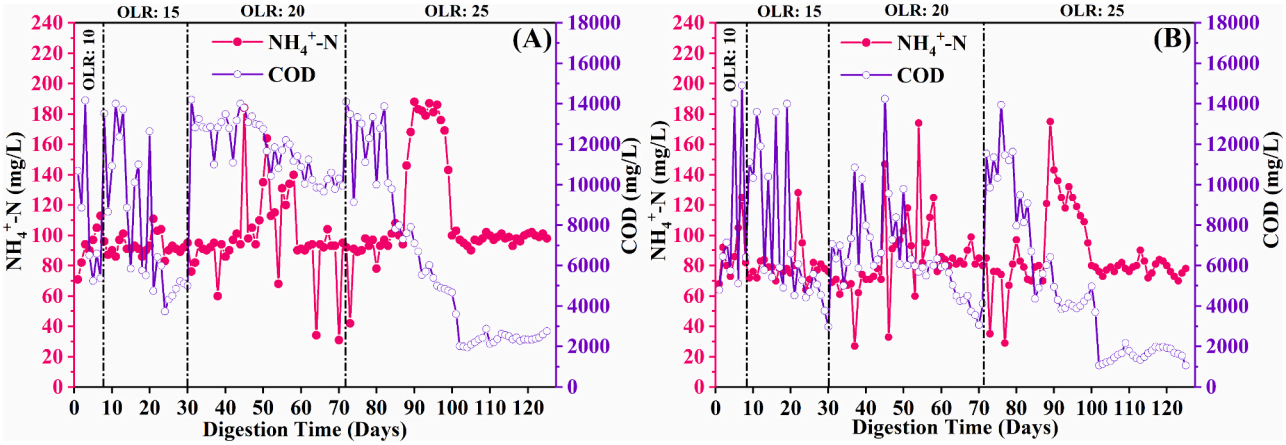


Fig. 3. Acidogenesis of food waste at different OLRs: (A) ammonia and COD concentrations without oxygen pretreatment, (B) ammonia and COD with oxygen pretreatment.

the release of a significant amount of NH_4^+-N. The effects of oxygen on NH_4^+-N and COD are shown in Fig. 3 (B). During the initial adaptation phase (Days 1–8), the low NH_4^+-N concentrations observed could be partly attributed to the increased initial degradation of nitrogenous compounds due to oxygen pretreatment, facilitating efficient nitrogen assimilation by the microbial community. With increased OLRs (days 9–71), fluctuations in NH_4^+-N levels suggest that oxygen pretreatment may increase initial nitrogen assimilation and that increasing OLRs impose stress on the microbial community, affecting its nitrogen processing capabilities. Elevated NH_4^+-N concentrations were observed at days 90–100, indicating a limitation in the nitrogen assimilation capacity of the microbial community despite the initial benefits of oxygen pretreatment.

The COD levels provide insight into the biodegradability of organic matter and the overall performance of the system in terms of VFAs production. Fig. 3 (A) shows that the initial low OLR phase (days 1–8) exhibited a high COD removal efficiency, which is indicative of effective organic matter degradation. As the OLR increased to 15 g/L and then to 20 g/L, the removal efficiency of COD decreased, suggesting that the microbial community could not effectively degrade the incoming organic matter. The last phase (days 72–125), with the highest OLR of 25 g/L, demonstrated a slight improvement in COD removal efficiency, possibly due to microbial adaptation to relatively high OLRs or the selection of more efficient degrading microbes. The effectiveness of oxygen pretreatment in enhancing organic matter degradation is reflected in the low COD levels in the early phase (Days 1–8), suggesting high degradation efficiency (Fig. 3B). With increased OLRs (days 9–71), despite the initial advantage provided by oxygen pretreatment, the increased COD concentrations indicate challenges in maintaining high degradation efficiency under higher OLRs (Fig. 3B). The complex pattern of COD levels, even at the highest OLR (Days 72–125), suggested that while oxygen pretreatment may facilitate initial organic matter degradation, the overall degradation efficiency of the system is influenced by the microbial community's adaptation to high OLRs and the inherent limitations of the process. Xu et al. reported that the degradation of aerobically pretreated FW was greater than that of the control, with hydrolysis in pretreated systems varying between 66.3 % and 78.1 %(Xu et al., 2014).

3.4. Microbial community analysis of food waste

3.4.1. Species diversity and richness

The findings of an investigation of the microbial diversity and richness in FW digestates, which are shown in Table 4, demonstrate the important influence that oxygen pretreatment has on the microbial communities that are involved in the biodegradation process. There were noticeable differences in the quantity of operational taxonomic units (OTUs) as well as the Shannon, ACE, and Chao1 indices between the oxygen-treated and nontreated groups. Understanding the complexity of microbial ecosystems requires the use of these metrics, which include the ACE and Chao1 indices, which estimate microbial richness, and the Shannon index, which represents diversity (Zhi et al., 2019). Compared with their oxygen-treated counterparts, the microbial communities were less rich and diverse on days 5 and 97 because of the absence of oxygen pretreatment, as evidenced by decreased OTU numbers and ACE and Chao1 indices. The ability of the microbial community to degrade FW is enhanced by oxygen pretreatment, which increases both the richness and diversity of microbes. The enhanced indicators (OTU number, ACE, and Chao1 indices) in samples treated with oxygen provide additional evidence of the beneficial effects of oxygen on microbial communities. These indicators point to a more resilient microbial ecosystem with effective hydrolysis and conversion processes. This microbial community enrichment and diversity, particularly in the oxygen-treated samples, highlights the crucial role that microbial dynamics play in maximizing waste-to-energy conversion processes. The bacterial coverage was 99 %, confirming the comprehensive sequencing depth and ensuring that the diversity within the samples was accurately captured and represented in the analysis.

3.4.2. Bacterial phyla

The composition of microbial communities at the phylum level in FW digestates was analyzed under different conditions: on day 5 and day 97,

Table 4
Indices indicating the microbial diversity and richness in food waste.

	Sample ID[a]	Seq num	OTU num	Shannon	ACE	Chao	Coverage rate
Bacteria	D5WOT	159707	706.0 ± 32.6	3.77 ± 0.1	830.0 ± 32.0	804.5 ± 26.8	0.998 ± 0.0
	D5WT	130122	715.7 ± 5.4	3.84 ± 0.0	850.9 ± 12.1	837.8 ± 9.8	0.998 ± 0.0
	D97WOT	127017	405.0 ± 35.7	2.85 ± 0.1	499.5 ± 35.9	484.2 ± 26.7	0.999 ± 0.0
	D97WT	154354	483.6 ± 31.6	3.12 ± 0.0	567.2 ± 30.4	548.1 ± 26.3	0.999 ± 0.0

[a] D5WOT: day 5 without pretreatment, D5WT: day 5 with pretreatment, D97WOT: day 97 without pretreatment, D97WT: day 97 with pretreatment.

both with and without oxygen pretreatment. The values indicate the relative abundance of different microbial phyla that play distinct roles in the degradation of FW and the production of VFAs. The results at the phylum level revealed that *Bacteroidota*, *Firmicutes*, *Synergistota*, and *Actinobacteria* were the most dominant phyla, as shown in Fig. 4. When FW was digested without oxygen, *Bacteroidota* and *Firmicutes* were the most abundant phyla, with relative abundances (RAs) of 66.1 % and 22.9 %, respectively, on day 5, as shown in Fig. 4A. After providing oxygen, the RA of *Firmicutes* (30.6 %) improved, whereas *Bacteroidota* (54.4 %) had a slightly lower abundance, as shown in Fig. 4B. On day 97 after pretreatment with oxygen, a complete shift in the microbial community was observed, as shown in Fig. 4D. The phylum-level shift in the microbial community structure from *Bacteroides* to *Firmicutes* suggested that FW in conjunction with oxygen increased the abundance of *Firmicutes*, which are known to hydrolyze proteins and carbohydrates to produce VFAs. This suggests that pretreatment may improve the hydrolysis process (Amin et al., 2021a). Additionally, a large percentage of bacteria in the *Firmicutes* phylum are facultative and grow at a faster pace in the presence of oxygen (Fu et al., 2015). Hydrolytic bacteria that aid in the breakdown of protein and sugar molecules into VFAs are also members of the phylum *Bacteroidota* (Maspolim et al., 2015), suggesting that hydrolytic bacterial enrichment is a major result of oxygen pretreatment. Understanding these microbial dynamics is crucial for optimizing the production of VFAs, which are valuable precursors for biofuel production and other biotechnological applications.

3.4.3. Bacterial genera and species

Fig. 5 shows the short-term (day 5) observations, in which without oxygen pretreatment, the community was diverse, with *Proteiniphilum* (27.1 %), *Petrimonas* (14.1 %), and *norank_f_Rikenellaceae* (9.3 %) being predominant. This diversity indicates that a dense anaerobic microbial ecosystem is capable of breaking down complex organic materials. The oxygen pretreatment led to a slight reduction in *Proteiniphilum* dominance to 21.5 %, a significant reduction in *Petrimonas* abundance to 3 %, and an increase in the abundance of *norank_f_Rikenellaceae* to 14.4 %. This indicates a shift toward microbes that can either tolerate or thrive in the presence of oxygen. *Proteiniphilum* contributes to the hydrolysis and fermentation of proteins and amino acids, although its direct role in

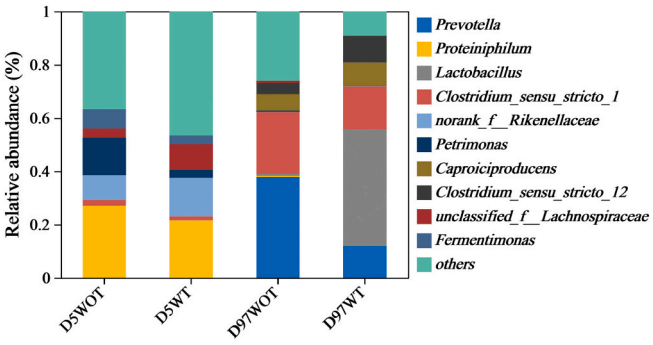


Fig. 5. Relative abundance of the dominant bacterial community in food waste at the genus level after pretreatment with and without oxygen for 5 and 97 days.

VFAs production might be limited compared with its role in breaking down complex organic matter (He et al., 2017). An increase in the abundance of the methane producer *norank_f_Rikenellaceae*, which is commonly found in anaerobic sludge, indicates an increase in methane production, which functions as a sort of rival process for the synthesis of medium-chain fatty acids. The presence of *Petrimonas* under anaerobic conditions suggests that it plays a role in the initial stages of organic matter breakdown, potentially facilitating the production of acetic acid in the presence of elemental sulfur (Wang et al., 2021).

Over the long term (day 97), without oxygen pretreatment, *Prevotella* (37.7 %) and *Clostridium* sensu *Stricto_1* (23.2 %) became dominant. The emergence of *Prevotella* plays a crucial role in the breakdown of proteins and carbohydrates, leading to the production of acetate and succinate, which are important VFAs in the context of FW degradation at low pH (Lebuhn et al., 2014; Xu et al., 2021). With oxygen pretreatment, a significant increase in *Lactobacillus* (43.6 %) was observed, indicating a shift toward lactic acid fermentation and other acid fermentation. It is involved mainly in lactic acid production but can also contribute to acetate and propionate production under certain conditions. Under aerobic conditions, its predominance indicates a transition toward lactic acid fermentation, which may compete with VFAs production pathways

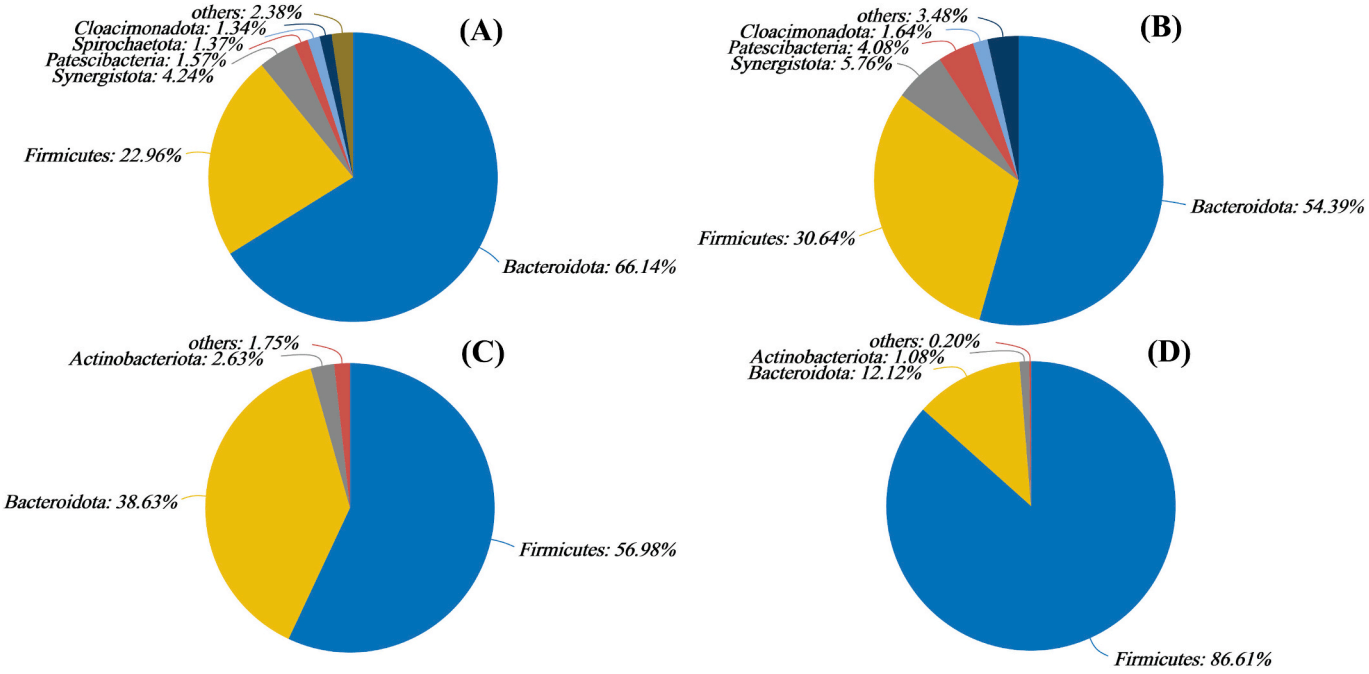


Fig. 4. Taxonomic classifications of the microbial community analysis of food waste with and without oxygen at the phylum level: (A) day 5 without pretreatment, (B) day 5 with pretreatment, (C) day 97 without pretreatment, (D) day 97 with pretreatment.

and use sugars to produce VFAs, CO_2, and D- and L-lactic acid. The optimal temperature for growth is 42 °C, with an acidic pH range of 3.0–6.5 (Boonkumklao et al., 2006; Niamsup et al., 2003). *Clostridium* sensu *Stricto_1,* which is significant (16.1 %) and is involved in the fermentation of sugars and amino acids to produce butyrate and other VFAs, can withstand acids and heat (Ma et al., 2020; Yang et al., 2019). The versatility and importance of VFAs production are highlighted by its presence under both aerobic and anaerobic conditions. The process dynamics and VFAs production profile may change dramatically over time, as evidenced by the shift in the makeup of the microbial community from a diversified anaerobic community in the short term to a more specialized community dominated by *Prevotella* and *Lactobacillus* in the long run. A broad variety of VFAs are anticipated to be produced under anaerobic circumstances, with the possibility of butyrate and acetate synthesis increasing with time. Conversely, because *Lactobacillus* is more prevalent in aerobic environments, there may be a decrease in

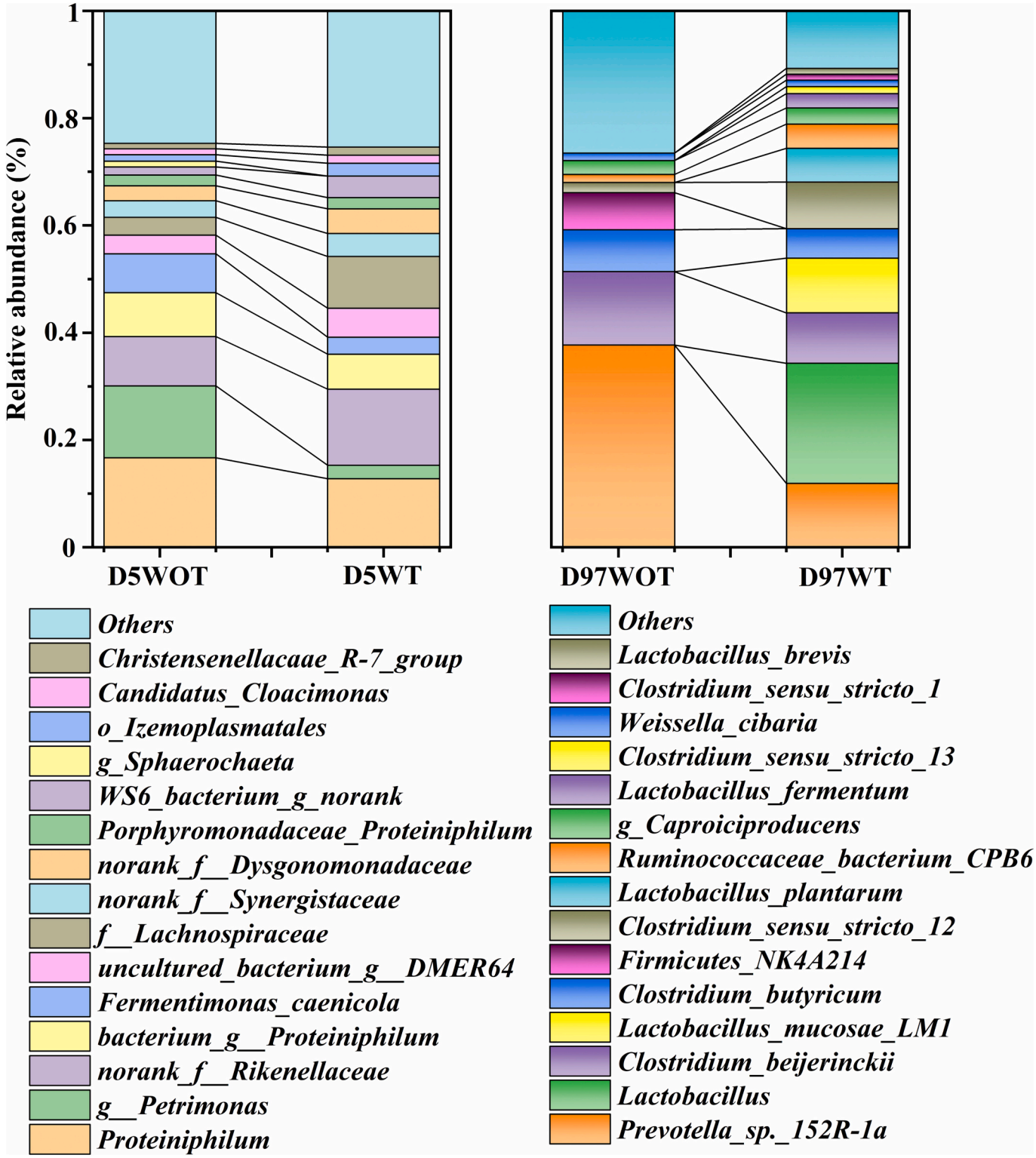


Fig. 6. Relative abundance of the dominant bacterial community in food waste at the species level after pretreatment with and without oxygen for 5 and 97 days.

VFA diversity and an increase in lactic acid production (Boonkumklao et al., 2006).

The bacterial community composition data for samples collected on day 5 with and without oxygen pretreatment are presented in Fig. 6, which reveals several important differences and implications for VFAs production from FW at the species level. A range of microbial species and their RA were examined in this study, which shed light on the interactions and possible functions of the microbial ecosystem in various environmental settings. *Proteiniphilum* (16.7 %) and *g_Petrimonas* (13.4 %) dominated the microbial community on day 5 without oxygen pretreatment, followed by *norank_f_Rikenellaceae* (9.2 %) and *bacterium_g_Proteiniphilum* (8.2 %). Other notable genera included *Fermentimonas_caenicola* (7.2 %) and a variety of other microbes, including uncultured_bacterium_g_DMER64, *f_Lachnospiraceae*, and *norank_f_Synergistaceae*. However, with oxygen pretreatment on day 5, the community structure shifted notably, with *norank_f_Rikenellaceae* becoming more dominant (14.2 %) and *Proteiniphilum* decreasing to 12.8 %. The abundance of *g_Petrimonas* significantly decreased to 2.5 %, suggesting sensitivity to oxygen. The abundance of *F_Lachnospiraceae* increased significantly to 9.6 %, and that of *uncultured_bacterium_g_DMER64* (5.4 %) and *norank_f_Synergistaceae* (4.3 %) increased. *Proteiniphilum* and *g_Petrimonas*, which are dominant in anaerobic samples, are known for their roles in the breakdown of complex organic substances, potentially leading to the production of acetate and propionate, key VFAs involved in waste degradation. Putrescine is converted into acetate, butyrate, molecular H_2, and ammonia by *Proteiniphilum*, which follows both acidogenic and acetogenic pathways (Matthies et al., 2000). *Petrimonas* belongs to the phylum *Bacteroidetes* and ferments mannose, maltose, galactose, arabinose, and glucose to produce acetate, CO_2, and H_2. It also ferments carbohydrates, proteins, and certain organic acids to produce acetate, CO_2, and H_2 (Huang et al., 2016). *Fermentimonas_caenicola* and *uncultured_bacterium_g_DMER64*, present under both conditions, contribute to the fermentation process and form acetic acid, propionic acid, CO_2 and traces of isovaleric acid (Hahnke et al., 2016). The increase in *f_Lachnospiraceae* under aerobic conditions indicated a shift toward a community that can contribute to the production of butyrate. To synthesize butyrate, metabolic intermediates that are necessary for the production of ketone bodies as energy sources are used (Kim et al., 2022). *Norank_f_Rikenellaceae* also increased during oxygen pretreatment, suggesting that it plays a role in the aerobic breakdown of organic matter, potentially affecting the VFA profile by favoring the production of specific acids to increase methane production at later stages (Yin and Wang, 2022).

Fig. 6 shows the microbial community composition data without oxygen pretreatment on day 97. *Prevotella* sp. *152R-1a* dominated at 37.7 %, suggesting a significant role in the fermentation process under anaerobic conditions (Lebuhn et al., 2014; Xu et al., 2021). *Clostridium beijerinckii* (a bacterium that initially produces organic acids) and *Clostridium butyricum* are also notable at 13.7 % and 7.8 %, respectively, indicating their importance in butyrate production (Fonseca et al., 2020; Mermejo et al., 2022). The abundances of *Firmicutes NK4A214* and *g_Caproiciproducens* were 6.9 % and 2.6 %, respectively, indicating diverse butyrate and caproate production potentials. The presence of *g_Caproiciproducens* indicated the potential for the production of caproate, a valuable chain-elongated VFAs that is mostly found in medium-chain carboxylate production systems (Esquivel-Elizondo et al., 2021). On the other hand, with oxygen pretreatment at 97 days (Fig. 6), *Lactobacillus* species became significantly more prevalent, with *Lactobacillus* at 22.4 %, *Lactobacillus mucosae LM1* at 10.2 %, and *Lactobacillus plantarum* at 6.3 %, highlighting a shift toward lactic acid fermentation. The increased prevalence of *Lactobacillus* species underlines a shift toward lactic acid production, which could compete with or complement VFAs production depending on the specific application (Boonkumklao et al., 2006). *Prevotella* sp. *152R-1a* was still present but was reduced to 11.9 %, suggesting its reduced dominance during oxygen pretreatment. Significant increases were observed in *Clostridium* sensu stricto *12* and *Ruminococcaceae bacterium CPB6,* which indicated a variety of microbial responses to oxygen availability. The prevalence of *Prevotella* and *Clostridium* species indicates that they have high potential for producing butyrate, acetate, and other short-chain fatty acids, all of which are necessary for the effective breakdown of FW (Ma et al., 2020; Xu et al., 2021). The variety of microbial species, such as *Ruminococcaceae_Bacterium_CPB6* and *Clostridium* sensu stricto, indicated a complicated network of pathways resulting in the formation of VFAs during oxygen pretreatment. The synthesis of caproic acid from lactate is effectively facilitated by *Ruminococcaceae_Bacterium_CPB6* (Tao et al., 2017). To optimize the AD process, a delicate balance of micronutrients is very important. Free metal ions, which include micronutrients such as calcium, magnesium, iron, nickel, and cobalt, have an impact on macronutrients. Enzymes need a limited quantity of metals as cofactors, which can disrupt the entire AD process. As previously indicated, certain enzymes, such as carbon monoxide dehydrogenase, use cobalt, an essential micronutrient, to produce methane from acetate. Various methanogenic strains contain copper as an adjuvant. Because of its low cost and conductive properties, iron is one of the most abundant elements. It can activate many enzymes, including pyruvate-ferredoxin oxidoreductase, and discharge extracellular polymers. For autotrophic methanogenesis to occur, nickel is necessary for molecules known as F factor 430 in cells, which are found in all methanogenic bacteria. Owing to their importance for enzymes found in acetoclastic methanogenesis pathways in the presence of vitamins, such as carbon monoxide dehydrogenase, acetyl-CoA decarboxylase, methyl-H 4 STP: HS-CoM methyltransferase, methyl-CoM reductase, and others, nickel, copper, and iron are the most frequently studied nutrient supplements in the AD process (Izadi et al., 2020).

The presence of anaerobic conditions creates a community of microbes that can ferment complex organics into a wide range of VFAs, which are crucial for the efficient degradation of FW. However, aerobic conditions seem to promote a different microbial community structure, possibly leading to a varied VFAs production profile, with potential shifts toward specific VFAs depending on the oxygen levels and microbial interactions. The shifts in microbial dominance and diversity between the two conditions suggest that environmental management, specifically oxygen regulation, could be leveraged to optimize VFAs production for bioenergy applications or as precursors in biochemical processes.

4. Conclusions and practical implications

This study demonstrates the significant potential of microaeration pretreatment in enhancing VFAs production from FW through AD. Key findings include the substantial increase in total VFAs yield, with a maximum of 7.4 g/L achieved through pretreatment. Microaeration altered the microbial community composition, favoring butyrate-producing *Firmicutes* and resulting in butyrate dominance (up to 92 % of total VFAs). Research has revealed how environmental factors, particularly oxygen availability, shape specialized microbial populations for tailored VFAs profiles. By combining microaeration pretreatment with organic loading optimization, this study presents a novel approach to improve and stabilize VFAs yields.

Some potential future implementations of this study that could be explored on the basis of butyric acid pathways are as follows:

1. As the study showed that butyrate production was favored under oxygen pretreatment conditions, further work could optimize pretreatment parameters such as the oxygen concentration and exposure time to maximize butyrate yields (Wang et al., 2024). Understanding key butyrate-producing microbial pathways, such as those of *Clostridium*, could aid in this understanding.
2. Microbes such as *Clostridium*, which dominate anaerobic conditions, could be isolated to study their butyric acid fermentation pathways

in detail. This could enable the development of defined cocultures for stable butyrate production.
3. Targeted enrichment of butyrate synthesis pathways in dominant microbes through techniques such as metagenomics could further improve butyrate titer and yields. Genes encoding butyryl-CoA: acetate CoA-transferase could be overexpressed (Tao et al., 2017).
4. Butyric acid has applications as a biofuel or chemical building block. Its derivatives, such as ethyl butyrate or butyl butyrate, can be produced *via* the esterification of butyric acid obtained from this process.
5. Controlled and efficient biomanufacturing of butyric acid and its derivatives can be achieved through synthetic microbial platforms that have been engineered with butyric acid pathways. This could realize the full potential of this process at an industrial scale.
6. The efficiency of waste valorization processes can be significantly increased in industrial settings by integrating oxygen pretreatment. Industries can customize VFAs profiles to fulfill individual needs for biochemical and biofuel production by optimizing the dynamics of the microbial community. Furthermore, by transforming FW into high-value goods, increased VFAs yields offer biorefineries important feedstocks, supporting the circular economy.

These findings provide a foundation for regulating anaerobic digestion at the microbiological level to achieve specific VFAs compositions, opening new avenues for maximizing biofuel and biochemical production through efficient waste-to-resource conversion methods. The potential for tailored VFAs production highlighted in this research has significant implications for various bioeconomic applications.

CRediT authorship contribution statement

Farrukh Raza Amin: Writing – original draft, Methodology. **Habiba Khalid:** Methodology. **Jingjing Wang:** Resources. **Yaxiang Li:** Validation. **Longxue Ma:** Visualization. **Wuxi Chen:** Conceptualization. **Yu Duan:** Investigation. **Yida Zhai:** Resources. **Demao Li:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgment

This project was financially supported by the Strategic Priority Research Program of the Chinese Academy of Sciences (XDC 0110300) and the Tianjin Synthetic Biotechnology Innovation Capacity Improvement Project (TSBICIP-IJCP-001-04, TSBICIP-BRFI-001, and TSBICIP-BRFI-002).

References

Aguirre-Garcia, J., 2019. Production and Recovery of Valuable Bioproducts through Anaerobic Fermentation of Organic Waste. University of Navarra.

Amin, F.R., Khalid, H., Zhang, H., Rahman, S., Zhang, R., Liu, G., Chen, C., 2017. Pretreatment methods of lignocellulosic biomass for anaerobic digestion. AMB Express 72, 1–12. https://doi.org/10.1186/s13568-017-0375-4.

Amin, F.R., Khalid, H., El-Mashad, H.M., Chen, C., Liu, G., Zhang, R., 2021a. Functions of bacteria and archaea participating in the bioconversion of organic waste for methane production. Sci. Total Environ. 763, 143007 https://doi.org/10.1016/j.scitotenv.2020.143007.

Amin, F.R., Khalid, H., Li, W., Chen, C., Liu, G., 2021b. Enhanced methane production and energy potential from rice straw by employing microaerobic pretreatment via anaerobic digestion. J. Clean. Prod. 296, 126434 https://doi.org/10.1016/j.jclepro.2021.126434.

APHA, 2005. APHA.: Standard Methods for the Examination of Water and Wastewater, 21st ed. American Public Health Association, Washington DC, USA.

Bao, N., Thi, D., Kumar, G., Lin, C., 2015. An overview of food waste management in developing countries : current status and future perspective. J. Environ. Manag. 157, 220–229. https://doi.org/10.1016/j.jenvman.2015.04.022.

Bastidas-Oyanedel, J.-R., Schmidt, J.E., 2019. Techno-Economic Analysis of Fermentation-Based Biorefinery: Creating Value from Food Residues. Biorefinery. Springer, in, pp. 535–552. https://doi.org/10.1007/978-3-030-10961-5.

Bhatt, A.H., Ren, Z.J., Tao, L., 2020. Value proposition of untapped wet wastes : Carboxylic acid production through anaerobic digestion. ISCIENCE 23, 101221. https://doi.org/10.1016/j.isci.2020.101221.

Boonkumklao, P., Kongthong, P., Assavanig, A., 2006. Acid and bile tolerance of lactobacillus thermotolerans, a novel species isolated from chicken feces. Agric. Nat. Resour. 40, 13–17.

Braguglia, C.M., Gallipoli, A., Gianico, A., Pagliaccia, P., 2018. Anaerobic bioconversion of food waste into energy : a critical review. Bioresour. Technol. 248, 37–56. https://doi.org/10.1016/j.biortech.2017.06.145.

Eryildiz, B., Taherzadeh, M.J., 2020. Effect of pH, substrate loading, oxygen, and methanogens inhibitors on volatile fatty acid (VFA) production from citrus waste by anaerobic digestion. Bioresour. Technol. 302, 122800 https://doi.org/10.1016/j.biortech.2020.122800.

Esquivel-Elizondo, S., Bagcı, C., Temovska, M., Jeon, B.S., Bessarab, I., Williams, R.B.H., Huson, D.H., Angenent, L.T., 2021. The isolate *Caproiciproducens sp*. 7D4C2 produces n-aproate at mildly acidic conditions from hexoses: Genome and rBOX comparison with related strains and chain-Elongating bacteria. Front. Microbiol. 11, 1–18. https://doi.org/10.3389/fmicb.2020.594524.

Fonseca, B.C., Bortolucci, J., da Silva, T.M., dos Passos, V.F., de Gouvêa, P.F., Dinamarco, T.M., Reginatto, V., 2020. Butyric acid as sole product from xylose fermentation by a non-solventogenic *Clostridium beijerinckii* strain under controlled pH and nutritional conditions. Bioresour. Technol. Reports 10, 1–10. https://doi.org/10.1016/j.biteb.2020.100426.

Fu, S., Shi, X., Dai, M., Guo, R., 2016. Effect of different mixed Microflora on the Performance of Thermophilic Microaerobic Pretreatment. Energy Fuel 30, 6413–6418. https://doi.org/10.1021/acs.energyfuels.6b00440.

Fu, S.F., He, S., Shi, X.S., Katukuri, N.R., Dai, M., Guo, R.-B., 2015. The chemical properties and microbial community characterization of the thermophilic microaerobic pretreatment process. Bioresour. Technol. 198, 497–502. https://doi.org/10.1016/j.biortech.2015.09.029.

Hahnke, S., Langer, T., Koeck, D.E., Klocke, M., 2016. Description of *Proteiniphilum saccharofermentans sp. nov*., *Petrimonas mucosa sp. nov*. and *Fermentimonas caenicola gen. Nov., sp. nov*., isolated from mesophilic laboratory-scale biogas reactors, and emended description of the genus *Proteiniphilum*. Int. J. Syst. Evol. Microbiol. 66, 1466–1475. https://doi.org/10.1099/ijsem.0.000902.

He, Q., Li, L., Zhao, X., Qu, L., Wu, D., Peng, X., 2017. Investigation of foaming causes in three mesophilic food waste digesters: reactor performance and microbial analysis. Sci. Rep. 7, 1–10. https://doi.org/10.1038/s41598-017-14258-3.

Huang, X., Mu, T., Shen, C., Lu, L., Liu, J., 2016. Effects of bio-surfactants combined with alkaline conditions on volatile fatty acid production and microbial community in the anaerobic fermentation of waste activated sludge. Int. Biodeterior. Biodegradation 114, 24–30. https://doi.org/10.1016/j.ibiod.2016.05.014.

Izadi, Parnian, Izadi, Parin, Eldyasti, A., Cheng, C., Beckley, M., 2020. Influence of vitamin coupled with micronutrient supplement on the biomethane production, process stability, and performance of mesophilic anaerobic digestion. Biomass Bioenergy 141, 105706. https://doi.org/10.1016/j.biombioe.2020.105706.

Jiang, J., Zhang, Y., Li, K., Wang, Q., Gong, C., Li, M., 2013. Volatile fatty acids production from food waste: Effects of pH, temperature, and organic loading rate. Bioresour. Technol. 143, 525–530. https://doi.org/10.1016/j.biortech.2013.06.025.

Jiang, L., Fu, H., Yang, H.K., Xu, W., Wang, J., 2018. Butyric acid: applications and recent advances in its bioproduction. Biotechnol. Adv. 36, 2101–2117. https://doi.org/10.1016/j.biotechadv.2018.09.005.

Jun, D., Yong-sheng, Z., Mei, H., Wei-hong, Z., 2009. Influence of alkalinity on the stabilization of municipal solid waste in anaerobic simulated bioreactor. J. Hazard. Mater. 163, 717–722. https://doi.org/10.1016/j.jhazmat.2008.07.066.

Kim, K., Lee, S., Park, S., Kim, N., Shin, C., Lee, S.K., Jung, Y., Yoon, D., 2022. Role of an unclassified *Lachnospiraceae* in the pathogenesis of type 2 diabetes: a longitudinal study of the urine microbiome and metabolites. Exp. Mol. Med. 54, 1125–1132. https://doi.org/10.1038/s12276-022-00816-x.

Kleerebezem, R., Joosse, B., Rozendal, R., Van Loosdrecht, M.C.M., 2015. Anaerobic digestion without biogas? Rev. Environ. Sci. Biotechnol. 14, 787–801. https://doi.org/10.1007/s11157-015-9374-6.

Komemoto, K., Lim, Y.G., Nagao, N., Onoue, Y., Niwa, C., Toda, T., 2009. Effect of temperature on VFA's and biogas production in anaerobic solubilization of food waste. Waste Manag. 29, 2950–2955. https://doi.org/10.1016/j.wasman.2009.07.011.

Ladero, M., Esteban, J., Bolívar, J.M., Santos, V.E., Martín-Domínguez, V., García-Martín, A., Lorente, Á., Escanciano, I.A., 2018. Food waste biorefinery for bioenergy and value added products. In: Sinharoy, A., Shu, L., Lens, P.N.L. (Eds.), Applied Environmental Science and Engineering for a Sustainable Future. Springer Nature Switzerland, pp. 185–224. https://doi.org/10.1002/adsu.201800109.

Lebuhn, M., Munk, B., Effenberger, M., 2014. Agricultural biogas production in Germany-from practice to microbiology basics. Energy Sustain. Soc. 4, 1–21.

Li, W., Siddhu, M.A.H., Amin, F.R., He, Y., Zhang, R., Liu, G., Chen, C., 2018. Methane production through anaerobic co-digestion of sheep dung and waste paper. Energy Convers. Manag. 156, 279–287. https://doi.org/10.1016/j.enconman.2017.08.002.

Ma, X., Yu, M., Yang, M., Zhang, S., Gao, M., Wu, C., Wang, Q., 2020. Effect of liquid digestate recirculation on the ethanol-type two-phase semi-continuous anaerobic digestion system of food waste. Bioresour. Technol. 313, 123534 https://doi.org/10.1016/j.biortech.2020.123534.

Mao, C., Wang, Y., Wang, X., Ren, G., Yuan, L., Feng, Y., 2019. Correlations between microbial community and C:N:P stoichiometry during the anaerobic digestion process. Energy 174, 687–695.

Maspolim, Y., Zhou, Y., Guo, C., Xiao, K., Ng, W.J., 2015. The effect of pH on solubilization of organic matter and microbial community structures in sludge fermentation. Bioresour. Technol. 190, 289–298. https://doi.org/10.1016/j.biortech.2015.04.087.

Matthies, C., Evers, S., Ludwig, W., Schink, B., 2000. *Anaerovorax odorimutans gen. Nov., sp. nov*., a putrescine-fermenting, strictly anaerobic bacterium. Int. J. Syst. Evol. Microbiol. 50, 1591–1594. https://doi.org/10.1099/00207713-50-4-1591.

Mermejo, B. da C., Bortolucci, J., de Andrade, A.R., Reginatto, V., 2022. The Non-solventogenic *Clostridium beijerinckii* Br21 Produces 1,3-Propanediol From Glycerol With Butyrate as the Main By-Product. Front. Sustain. Food Syst. 6, 3–5. https://doi.org/10.3389/fsufs.2022.848022.

Niamsup, P., Sujaya, I.N., Tanaka, M., Sone, T., Hanada, S., Kamagata, Y., Lumyong, S., Assavanig, A., Asano, K., Tomita, F., Yokota, A., 2003. *Lactobacillus thermotolerans sp. nov*., a novel thermotolerant species isolated from chicken faeces. Int. J. Syst. Evol. Microbiol. 53, 263–268. https://doi.org/10.1099/ijs.0.02347-0.

Ren, Y., Yu, M., Wu, C., Wang, Q., Gao, M., Huang, Q., Liu, Y., 2018. A comprehensive review on food waste anaerobic digestion: Research updates and tendencies. Bioresour. Technol. 247, 1069–1076. https://doi.org/10.1016/j.biortech.2017.09.109.

Selder, L., Turunen, R., Vuoristo, K.S., Uusitalo, J., Zeng, A.P., 2021. Volatile and medium chain fatty acid production from cabbage residues using mixed culture fermentations and in-situ product removal by electrodialysis. Bioresour. Technol. Reports 16, 100828. https://doi.org/10.1016/j.biteb.2021.100828.

Shen, D., Yin, J., Yu, X., Wang, M., Long, Y., Shentu, J., Chen, T., 2017. Acidogenic fermentation characteristics of different types of protein-rich substrates in food waste to produce volatile fatty acids. Bioresour. Technol. 227, 125–132. https://doi.org/10.1016/j.biortech.2016.12.048.

Tao, Y., Zhu, X., Wang, H., Wang, Y., Li, X., Jin, H., Rui, J., 2017. Complete genome sequence of *Ruminococcaceae* bacterium CPB6: a newly isolated culture for efficient n-caproic acid production from lactate. J. Biotechnol. 259, 91–94.

Tenci, N.A., Ammam, F., Huang, W.E., Thompson, I.P., 2023. Anaerobic co-digestion of Euphorbia tirucalli with pig blood for volatile fatty acid production. Bioresour. Technol. Reports 21, 101333. https://doi.org/10.1016/j.biteb.2023.101333.

Uddin, M.M., Wright, M.M., 2023. Anaerobic digestion fundamentals, challenges, and technological advances. Phys. Sci. Rev. 8, 2819–2837. https://doi.org/10.1515/psr-2021-0068.

Viancelli, A., Schneider, T.M., Demczuk, T., Delmoral, A.P.G., Petry, B., Collato, M.M., Michelon, W., 2023. Unlocking the value of biomass: Exploring microbial strategies for biogas and volatile fatty acids generation. Bioresour. Technol. Reports 23, 101552. https://doi.org/10.1016/j.biteb.2023.101552.

Wagner, A.O., Lackner, N., Mutschlechner, M., Prem, E.M., Markt, R., Illmer, P., 2018. Biological pretreatment strategies for second-generation lignocellulosic resources to enhance biogas production. Energies 11, 1–14. https://doi.org/10.3390/en11071797.

Wainaina, S., Lukitawesa, Awasthi, M.K., Taherzadeh, Mohammad, J., 2019. Bioengineering of anaerobic digestion for volatile fatty acids, hydrogen or methane production: a critical review. Bioengineered 10, 437–458. https://doi.org/10.1080/21655979.2019.1673937.

Wang, L., Jin, W., Cai, F., Song, C., Jin, Y., Liu, G., Chen, C., 2024. Performance and mechanism of various microaerobic pretreatments on anaerobic digestion of tobacco straw. Bioresour. Technol. 393, 130092 https://doi.org/10.1016/j.biortech.2023.130092.

Wang, P., Tan, Z., Guan, L., Tang, S., Zhou, C., Han, X., Kang, J., He, Z., 2015. Ammonia and amino acids modulates enzymes associated with ammonia assimilation pathway by ruminal microbiota in vitro. Livest. Sci. 178, 130–139. https://doi.org/10.1016/j.livsci.2015.05.033.

Wang, Q., Cole, J.R., 2024. Updated RDP taxonomy and RDP classifier for more accurate taxonomic classification. Microbiol. Resour. Announc. 13, 2–4. https://doi.org/10.1128/mra.01063-23.

Wang, Q., Li, H., Feng, K., Liu, J., 2020. Oriented fermentation of food waste towards high-value products: a review. Energies 13, 1–29. https://doi.org/10.3390/en13215638.

Wang, Y., Wei, W., Wu, W., Sun, J., Xu, Q., Wang, D., Song, L., Ni, B., 2021. Improving medium-chain fatty acid production from anaerobic fermentation of waste activated sludge using free ammonia. ACS Environ. Sci. Technol. Eng. 1, 478–489. https://doi.org/10.1021/acsestengg.0c00179.

Xu, H., Li, Y., Hua, D., Zhao, Y., Chen, L., Zhou, L., Chen, G., 2021. Effect of microaerobic microbial pretreatment on anaerobic digestion of a lignocellulosic substrate under controlled pH conditions. Bioresour. Technol. 328, 124852 https://doi.org/10.1016/j.biortech.2021.124852.

Xu, S., Selvam, A., Wong, J.W.C., 2014. Optimization of micro-aeration intensity in acidogenic reactor of a two-phase anaerobic digester treating food waste. Waste Manag. 34, 363–369. https://doi.org/10.1016/j.wasman.2013.10.038.

Yang, G., Yin, Y., Wang, J., 2019. Microbial community diversity during fermentative hydrogen production inoculating various pretreated cultures. Int. J. Hydrog. Energy 44, 13147–13156. https://doi.org/10.1016/j.ijhydene.2019.03.216.

Yin, Y., Wang, J., 2022. Medium-chain carboxylates production by co-fermentation of sewage sludge and macroalgae. Bioresour. Technol. 347, 126718 https://doi.org/10.1016/j.biortech.2022.126718.

Zhang, H., Ning, Z., Khalid, H., Zhang, R., Liu, G., Chen, C., 2018. Enhancement of methane production from cotton stalk using different pretreatment techniques. Sci. Rep. 8, 1–9. https://doi.org/10.1038/s41598-018-21413-x.

Zhi, S., Li, Q., Yang, F., Yang, Z., Zhang, K., 2019. How methane yield, crucial parameters and microbial communities respond to the stimulating effect of antibiotics during high solid anaerobic digestion. Bioresour. Technol. 283, 286–296. https://doi.org/10.1016/j.biortech.2019.03.083.